

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (IT)

May 2014

IT206 Object Oriented Programming using Java

Date: 03.05.2014, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION - I

Q - 1 Attempt following.

- (a) Explain basic features for java programming language. [05]
- (b) Explain bitwise unsigned right shift operator with example. [02]

Q - 2 Answer the following.

- (a) Write a java program that creates a class named Student which have instance variables name, age and percentage. These should be of types String, int and float respectively. Create an object of 5 students and write methods to set and display its instance variables. [05]
- (b) **Attempt Following.** [09]
 - 1. Describe the use of static keyword with example.
 - 2. What is interface? Differentiate between class and interface.
 - 3. What is constructor? Explain different types of constructors.

OR

Q - 2 Answer the following.

- (a) Write a java program to read two numbers as a command line argument. And find out largest number out of two numbers. [05]
- (b) **Attempt Following** [09]
 - 1. Describe the use of this keyword with example.
 - 2. Explain method overloading with example.
 - 3. What is package? Explain steps to create package with example.

Q - 3 Attempt following.

- (a) What is Thread? Draw and explain life cycle of Thread. [07]
- (b) Write an applet program which performs following. [05]
Initiate applet window having one button labeled START. When START button is clicked label of that button is changed from START to STOP. When label of the button is STOP, and if button is clicked label of that button is changed from STOP to START.
- (c) What is an action event? [02]

OR

Q - 3 Attempt following.

- (a) What is applet? Draw and explain life cycle of Applet. [07]
- (b) Write a java program using the concepts of File streams. One NUMBER.TXT file contains set of numbers. Find out how many odd and even numbers are present in that file. [05]
- (c) Name the six condition that generate an exception. [02]

SECTION – II

Q - 4 Attempt following.

- (a) What is an exception? Explain five keywords which are used to generate and handle an exception in java. [07]

Q - 5 Attempt following.

- (a) What is generalization? Explain three purpose of generalization in object oriented development with an example. [06]
(b) Define state diagram. Draw state diagram for "Phone Line System". [08]

OR

Q - 5 Attempt following.

- (a) Explain object oriented methodology in detail. [06]
(b) Define object and class. Draw class diagram for "Ticket Reservation System". [08]

Q - 6 Attempt following.

- (a) Draw and explain the notations of DFD. [04]
(b) Define scenario. Write the scenario for traffic signal system. [04]

OR

- (b) Draw an activity diagram for "processing an order". [04]
(c) **Differentiate Following. (Any Three)** [06]
1. procedural oriented programming and object oriented programming
2. state chart diagram and activity diagram
3. aggregation and association
4. sequence diagram and collaboration diagram
